

Debangana Ghosh

Mumbai, MH • 84339 55740 • ranjana1997@gmail.com
portfolio-dghosh.vercel.app • github.com/orackle • in/ghosh-debangana

SUMMARY

Software engineer with experience in applied **machine learning**, **NLP**, and **Generative AI**. Built and deployed ML-driven systems including sentiment models, RAG pipelines. Focus on scalable backend design, observability and deploying **Generative AI solutions**.

EDUCATION

SUNY, Binghamton University

Master of Science in Computer Science

Aug 2022 – May 2024

University of Alberta

Bachelor of Science in Computer Science

Sept 2017 – May 2021

SKILLS

Generative AI / LLM: RAG, Prompt Engineering, Vector Embeddings, FAISS, GPT model integration, Langchain

Machine Learning: TensorFlow, PyTorch, scikit-learn, NLP, sentence-transformers

Languages: Python, SQL, Java, Go

Backend / Infrastructure: FastAPI, Django, Flask, Node.js, REST APIs, PostgreSQL, MongoDB, Kafka

Cloud / MLOps: AWS, GCP, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD

PROFESSIONAL EXPERIENCE

Binghamton University, Software Engineer | Binghamton, NY

Jun 2024 – Jul 2025

- **Project 1:** Built and deployed NLP models (TensorFlow, FastAPI) for real-time **sentiment analysis**; applied data augmentation (10K+ adversarial samples) to address class imbalance, improving classification performance.
- **Project 2:** Built a **smart documentation tool** that utilizes **Generative AI** to instantly categorize and index user inputs; deployed via **containerized microservices (Docker)** to support real-time search and reduce manual data entry.
- **Project 3:** Built **automated Python pipelines** to clean and normalize multi-source vendor data, handling inconsistent schemas and malformed records using mapping tables and validation rules.
- **Project 4:** Integrated **OpenAI Whisper** and **GPT models** to transcribe and classify raw audio into structured clinical summaries, dynamically generating context-aware follow-up questions to streamline patient intake and documentation.

Information Technology Services, Software Engineer | Binghamton, NY

May 2023 – Jun 2024

- **Project 1:** Built **Python REST APIs** handling **16K+ daily requests**; improved latency & reliability for multiple university applications.
- **Project 2:** Designed and developed a **service-status monitoring system** (Python, JS, Oracle); that slashed incident response times and operational overhead by **80%** through automated alerting.
- **Project 3:** Migrated and automated **onboarding workflows** for 200+ employees; reduced manual processing and errors.
- **Project 4:** Developed a high-precision **network traffic analyzer** using Python and Shell, boosting classification accuracy across 100+ distributed endpoints.

Ada's Team & Areto Labs, Software Engineer Internships | Edmonton, Alberta

Aug 2019 – Nov 2021

- **Project 1:** Developed a **Django + React chatbot** serving **100K+ daily queries**, applying **NLP** for entity extraction and context-aware responses.
- **Project 2:** Rebuilt a client platform with **React**, **Tailwind**, and **MongoDB**, onboarding 300+ users with responsive, accessible UX.
- **Project 3:** Implemented **CI/CD pipelines** with **GitHub Actions** and containerized services using **Docker/Kubernetes**, cutting deployment time by **50%**.
- **Project 4:** Automated **GCP backups** and optimized **PostgreSQL schema**, improving reliability and query performance.

PROJECTS

RAG Q&A pipeline — LangChain, Python, Sentence-Transformers

Dec 2025

- Built a Retrieval-Augmented Generation system combining sentence-transformer embeddings, a vector store, and an LLM to answer natural-language questions grounded in a custom document set.

Subreddit Toxicity Analysis with NLP — Flask, PostgreSQL, Bash, APIs

Jun 2025

- Built a social media trend analysis platform using Reddit & TMDB APIs, sentiment analysis, and Flask to identify trending movies/TV shows and measure toxicity across genres.

CERTIFICATIONS

• **Data Analysis with Python** — IBM Skillsbuild

July 2026

• **Machine Learning with Python** — IBM Skillsbuild

Apr 2026

• **Introduction to Generative AI** — Google

Aug 2025